3. What is the output of the following pro

```
#include <stdio.h>
int fred (int x, int *y)
{
        int z;
        z = x + 3;
        *y = z + x;
        return (z);
}
int
main (void)
{
        int n=10, a;
        n = fred (n, &a);
        printf ("%d", n+a);
        return (0);
}
```

a) 36 b) 20 c) 16

e) none of these

4. What is the output of the following progra

```
#include <stdio.h>
int fred (int y)
{
        y = y * 2;
        return (y);
}
```

5. In C, if an array is declared as **double array [100]**, what can be substituted for **array** *after the declaration*?

a) array[0]
b) &array[0]
c) array[0][0]
d) *array
e) none of these

6. In C, if an array is declared as **char array [12]** to hold a word, how long is the longest word it can hold?

a) 12 letters
b) 11 letters
c) 13 letters
d) 12 words
e) none of these

7. What is the output of the following statement?
printf ("%4.2lf", atof ("47f3"));

a) 47
b) 47.00
c) 47.30
d) 47300.00
e) none of these

8. In C, the main program header when using command line arguments is **int main (int argc, char* argv[])**. What is inside argc?

a) the arguments themselves
b) the number of characters in the arguments
c) the number of arguments
d) a string describing the arguments
e) none of the above

9. *If a function header (prototype) is:*
void pebbles (int x[], int y, int *z). Identify the statement that *can be used inside this function without causing a compile-time error.*

a) x = pebbles [i];
b) z = x;
c) z = *y + 1;
d) return (x[]);
e) they can all be used

10. What is the output of the following program?

```
#include <stdio.h>

typedef struct road_trip
{
    char place;
    struct road_trip* next;
}trip;


int
main (void)
{
        trip s1, s2, s3, s4, s5, s6;
        trip *ptr;

        s1.place = 'A';
        s2.place = 'B';
        s3.place = 'C';
        s4.place = 'D';
        s5.place = 'E';
        s5.next = &s1;
        ptr = &s2;
        s1.next = &s4;
        s3.next = NULL;
        s4.next = &s3;
        s2.next = &s5;
        while (ptr != NULL)
        {
                printf ("%c  ", ptr -> place);
                ptr = ptr -> next;
        }
        return (0);
}
```

S1 s2 s3
A B C

S1 s2
&s4 &s5
D E

a) A B C D E

b) C D A E B

```
{
        trip s1, s2, s3, s4, s5, s6;
        trip *ptr;

        s1.place = 'A';
        s2.place = 'B';
        s3.place = 'C';
        s4.place = 'D';
        s5.place = 'E';
        s5.next = &s1;
        ptr = &s2;
        s1.next = &s4;
        s3.next = NULL;
        s4.next = &s3;
        s2.next = &s5;
        while (ptr != NULL)
        {
                printf ("%c  ", ptr -> place);
                ptr = ptr -> next;
        }
        return (0);
}
```

S1 S2 S3
A B C

S1 S2 S3
D E

a) A B C D E
b) C D A E B
c) B E A D C
d) E D C B A
e) None of these

11. What is the output of the following program?

```
#include <stdio.h>
int
main (void)
{
    double* array; int i;
    array = (double *) calloc (3, sizeof(double));
    for (i = 0; i < 3; ++i)
        array[i] = array[i] + i;
    printf ("%3.1lf", array[2]);
    free (array); return (0);
}
```

a) 0.0 b) 1.0 c) 2.0 d) 3.0
e) a garbage value

Professors and invigilators will not answer any questions during the exam (except if you need help understanding an English word)

SECTION II (1 mark per question):
Identify the correct missing line in the following C programs. The missing line must make the program produce results (no program error) consistent with the labelling in the output statements. The position of the missing line is indicated by a ▶ in the left margin.

12.

```
#include <stdio.h>
#include <string.h>
typedef struct car {
        char color[20];
        int year;
        char make[20];
}car;
int
main (void)
{
        car x = {"yellow", 2009, "Volkswagen"};
        car* ptr;
        char brand[20];
        ptr = &x;
▶
        printf ("The make of the car is %s", brand);
        return (0);
}
```

ptr = ptr -> next

a) brand = x.make;
b) strcpy (brand, ptr->make);
c) strcpy (brand, ptr.make);
d) strcpy (brand, x->make);
e) none of these

13.

```
#include <stdio.h>
int
main (void)
{
▶
        x[0][0] = 50.0;
        printf ("The temperature is %lf", x[0][0]);
        return (0);
}
```

a) double x;
b) double x[10];
c) double* x[10][10];
d) double x[10][10];
e) none of these

14.

```
#include <stdio.h>
void barney (int* x, int y[])
{
        *x = y[0];
}
int
main (void)
{
        int array[3] = {1,2,3}, n;
        /* prints out the first element (cell) of the array */

        printf("%d\n", n);
        return (0);
}
```

a) n = y[0];
b) n = array[1];
c) n = barney (x, array);
d) barney (&n, array);
e) barney (n, array[0]);

15.

```
#include <stdio.h>
int dino (int y[])
{
        int x;
        x = y[0];
        return (x);
}
int
main (void)
{
        int array[3] = {1,2,3}, n;
        /* prints out the first element (cell) of the array */

        printf("%d\n", n);
        return (0);
}
```

a) n = y[0];
b) n = array[1];
c) n = dino (array);
d) dino (&n, array);
e) dino (n, array[0]);